PT7004 PRACTICAL:

NEURO: Electrotherapy and Pain Management (TENS)

Date: Tue 25/02/2025 **Time: 14:00 -17:00**

Learning Outcomes:

- Explain indications, contraindications and safety measures required when using TENS.
- Apply different therapeutic frequencies, pulsing, intensity, and durations.
- Demonstrate clinical reasoning for the use of ultrasound
- Application of NMES including pulses, durations and amplitudes for strengthening, re-education, spasm reduction, oedema Reduction
- Demonstrate clinical reasoning for the use of NMES

Session Outline:

Electrical stimulation is one of the treatment options available to us as therapists. The following are the more uses of electrical stimulation:

- Pain Relief (as utilised in Transcutaneous Electrical Nerve Stimulation (TENS))
- Muscle contraction/strengthening (NMES)

		Timings
1	Safety Checks	5 mins
2	Contraindications	5 mins
3	Demonstration of TENS	15 mins
4	Student use of TENS	20 mins
5	NMES demonstration	15 mins
6	Student use of NMES	15 mins

A. PAIN RELIEF (TENs) (45 mins)

Tutor to demonstrate the application of TENS for pain relief (15 mins)

Procedure:

1. Collect the necessary equipment: electrical stimulator; electrodes (one pair of self-adhesive electrodes) and cable leads; adhesive tape; soap and water for cleaning the skin.

   Equipment:

   A battery or mains operated TENS stimulator should be used:

   Biphasic, asymmetrical, square wave with zero net DC component, amplitude 0-50mA; pulse frequency 1-150; pulse width 0.05- 02 msec.

2. Conduct general safety checks with respect to the equipment.

Safety Checks

- Mechanical safety of equipment (leads, electrodes), service records etc.
- Output of the machine.
- Check subject with respect to contraindications listed below:

Contraindications:

- Pacemakers or areas where it could cause malfunction of any electronic devices
- to anterior neck or carotid sinus
- to areas of impaired peripheral circulation
- to persons with active deep vein thrombosis or thrombophlebitis deep vein thrombosis
- areas with impaired sensation
- pelvis, abdomen, trunk and low back during early pregnancy (1-3 months)
- regions of known or suspected malignancy
- to the chest in persons with cardiac disease, arrhythmias, or heart failure
- to the neck or head region of persons known to have seizures
- to recently radiated tissues
- to damaged or at-risk skin areas that would result in uneven conduction of current (excluding
  open wounds where the specific intent is to use electrical stimulation for tissue healing)

3. Explain the treatment fully to the subject.
4. Position the subject so that they are comfortable. Ensure that the position allows the electrodes to be placed over the necessary contact points. The skin should be uncovered and examined for any contraindications to treatment.
5. Wash the skin to ensure good connection between skin and electrode.

6. Placement of electrodes: This varies in practice according to the cause and location of the pain.
   TENs electrodes can be placed at the following positions:

   - above and below or either side of 'a painful area'
   - over trigger points
   - over a nerve where it is superficial, e.g. the peroneal nerve
   - proximal to the point of pain if part not accessible due to e.g. a cast or over relevant dermatome
   - over a nerve trunk

   ENSURE THAT THE SUBJECT CAN MOVE ABOUT WITH TREATMENT IN SITU - REMEMBER THAT THE EQUIPMENT MAY BE USED FOR UP TO 24 HOURS AT A TIME!

7. Explain what the treatment will feel like (this depends on the pulse frequency)
   - slow rate (around 10 Hz) - slow tickling sensation
   - high rate (100 Hz +) - continuous buzzing sensation

Warnings: no specific warnings are necessary but ensure the subject is clear about what to expect; they will experience some sensory stimulation which is exacerbated when the active electrode is not well placed or when there is poor contact between the skin / contact medium / electrode.

8. Always begin with the intensity setting at zero. Increase the output until the subject is aware of a mild buzzing or pulsing. Intensity will be dependent on whether convention, low or burst TENS is desired.
9. Scan through the full range of pulse frequencies. Note any differences in sensation. (Client will be able to identify which is most effective in relieving their pain).
10. If, after a short time, the subject stops feeling the stimulus the intensity/frequency/pulse duration may be tweaked.

<u>THE INTENSITY SHOULD NEVER BE STRONG OR PAINFUL</u>

11. You may need to move the electrode around a little to find the optimum position for stimulation. This is because surface anatomy may vary a little between subjects. 12. NEVER LIFT THE ACTIVE ELECTRODE FROM THE SKIN OR REPLACE IT WITHOUT TURNING THE INTENSITY TO ZERO.

13. Terminate the treatment and check the skin condition.
14. Keep a full record of your treatment, noting:
    - Area of body treated
    - Patient positioning
    - Specific stimulation parameters

- Electrode placement
- Treatment duration
- Patient's response to treatment

Hazards:

- local skin irritation at electrode sites due to prolonged use of equipment
- allergy to gels or adhesives
- muscle damage or joint injury if the intensity of the current and therefore contraction is too strong
- cardiac arrhythmia's due to autonomic stimulation may occur when stimulating in the chest region
- electrical shock
- burns

Students Activity 1 (20 mins)

Students practice delivery of TENs for hemiplegic shoulder pain

NEUROMUSCULAR ELECTRICAL STIMULATION (NMES)

Recommended parameter settings for NMES

Goal	Pulse frequency	Duration	Amplitude	On: off	Ramp
Strengthening	35-80Hz	150-200µs	at least > 10% of injured MVIC; >50% MVIC of uninjured	6-10s on; 50-120s off	At least 2s
Re-education	35-50Hz	150-200µs	To achieve functional activity	Depending on functional task	At least 2s
Spasm reduction	35-50Hz	150-200µs	To achieve visible contraction	2-5s on; 2-5s off	At least 1s
Oedema Reduction	35-50Hz	150-200µs	To achieve visible contraction	2-5s on; 2-5s off	At least 1s

Tutor to demonstrate the use of NMES for muscle strengthening in neurology (15 mins)

1. Collect the necessary equipment: electrical stimulator; electrodes and cable leads; contact gel may be required; adhesive tape; soap and water for cleaning the skin.

2. Conduct general safety checks with respect to the equipment.

Safety Checks

- Mechanical safety of equipment.
- Output of the machine.
- Test the equipment on yourself in front of the subject. Use the thenar muscles of the hand and place one electrode at each end of the muscle bulk. Demonstrate the treatment and note the approximate intensity of current required to produce a contraction. The last will vary to some degree depending on the muscle being activated.

3. Position the subject so that he/she is comfortable and can remain in the given position for the duration of the treatment. Ensure that the position allows the electrode to be placed over the necessary motor points and that the limb can be secured to prevent the motion through the range of the joint that the muscle crosses. The skin should be uncovered and examined for any contraindications to treatment.

4. Check subject with respect to contraindications listed below:

Contraindications:

- Pacemakers or areas where it could cause malfunction of any electronic devices
- to anterior neck or carotid sinus
- to areas of impaired peripheral circulation
- to persons with active deep vein thrombosis or thrombophlebitis deep vein thrombosis
- check skin sensation for information only - lack of sensitivity is NOT a contraindication to treatment but you should take particular care to watch for any changes in skin colour due to chemical irritation of the skin under the electrodes
- pelvis, abdomen, trunk and low back during early pregnancy (1-3 months)
- regions of known or suspected malignancy
- to the chest in persons with cardiac disease, arrhythmias, or heart failure

- to the neck or head region of persons known to have seizures
- to recently radiated tissues
- to damaged or at-risk skin areas that would result in uneven conduction of current (excluding
- open wounds where the specific intent is to use electrical stimulation for tissue healing)

5. Wash the skin over the region of electrode contact. You may reduce skin resistance by soaking the skin for three to four minutes either in a bath or with a warm damp pad.
6. **Stimulation of muscle via nerve:** Place an electrode more centrally on the part to be treated or over the spine and another electrode over the motor point. Apply gentle pressure - poor contact will lead to increased sensory irritation and the need for increased current. You may need to move the electrode around a little to find the optimum position for stimulation. This is because the position of motor points varies a little between subjects.
7. **Stimulation of muscle directly:** Place the electrode at either end of the bulk to be stimulated
8. Restrict the limb to move in order to elicit an isometric contraction.

Selecting stimulation parameters: – frequency; pulse duration; on: off time; ramp time.

Warnings: no specific warnings are necessary but ensure the subject is clear about what to expect; they will experience some sensory stimulation which is exacerbated when the active electrode is not well placed or when there is poor contact between the skin / contact medium / electrode.

9. Increase the intensity until a muscle contraction is produced. The duration and frequency should mimic a normal, active muscle contraction

10. NEVER LIFT THE ACTIVE ELECTRODE FROM THE SKIN OR REPLACE IT WITHOUT TURNING THE INTENSITY TO ZERO. You may roll the electrode on the skin provided you maintain good contact with the tissue.
11. Terminate the treatment; check the skin condition.
12. Keep a full record of your treatment.

<u>Students Activity 2 (15 mins)</u>

Demonstrate the application of NMES for muscle strengthening